

Surface EMG and acceleration signals in Parkinson's disease: feature extraction and cluster analysis

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Abstract We present an advanced method for feature extraction and cluster analysis of surface electromyograms (EMG) and acceleration signals in Parkinson's disease (PD). In the method, 12 different EMG and acceleration signal features are extracted and used to form high-dimensional feature vectors. The dimensionality of these vectors is then reduced by using the principal component approach. Finally, the cluster analysis of feature vectors is performed in a low-dimensional eigenspace. The method was tested with EMG and acceleration data of 42 patients with PD and 59 healthy controls. The obtained discrimination between patients and controls was promising. According to clustering results, one cluster contained 90% of the healthy controls and two other clusters 76% of the patients. Seven patients with severe motor dysfunctions were distinguished in one of the patient clusters. In the

future, the clinical value of the method should be further evaluated.

Keywords Electromyography (EMG) · Parkinson's disease · Nonlinear analysis · Signal morphology · Cluster analysis

1 Introduction

The neuromuscular system of humans can be studied by measuring the electrical potential it elicits in a muscular contraction. This electrical potential is called the electromyogram (EMG) and recording of it from the skin surface EMG measurement. Surface EMG signals are often analyzed using amplitude and spectral based methods. These methods are used to measure the degree of muscle activation and fatigue. The neuromuscular function can be measured indirectly by measuring limb movements controlled by it. This can be done by using movement sensors such as accelerometers.

Parkinson's disease (PD) is a common neurodegenerative disorder in which the control of skeletal muscles is disturbed. This results from a progressive dopaminergic neuronal loss in the substantia nigra in the brain and leads to the typical symptoms of PD. These symptoms are involuntary and oscillatory movements (tremor), slowness of movements (bradykinesia), increased muscle tone (rigidity), and postural instability [20]. According to current diagnostic criteria [2], patients are considered to have PD when they have two of these four characteristic symptoms and the potential causes of secondary Parkinsonism have been excluded. The asymmetric onset of symptoms and a good response to PD medication are important in discriminating PD from other diagnoses.

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Diagnosis of PD is primarily based on a subjective clinical evaluation of symptoms. The evaluation of PD symptoms can be done by using standardized rating scales such as the Unified Parkinson's Disease Rating Scale (UPDRS). There are no objective and quantitative methods in common use to assess the abnormal muscle control in PD. PD diagnosis is highly uncertain in the early stages and up to 30% of patients diagnosed with PD are later clinically reclassified [17]. Because of these difficulties, it is important to examine the potentiality of neurophysiologic measurements such as EMG in helping the diagnosis. EMG and movement measurements would be objective methods to assess neuromuscular function in PD. However, they are not used in the clinical assessment of PD at the moment.

The EMGs of patients with PD differ from the EMGs of healthy persons and some mathematical approaches have been proposed to assess these differences. The main features of PD-like EMG signals are an increased tonic background activity and an alternating pattern of EMG bursts. This characteristic pattern of EMG bursts is thought to result from the centrally induced motor unit (MU) synchronization [20]. The mathematical approaches used for EMG analysis in PD include spectral based methods (mean frequency, proportion of power at particular frequency bands) [5, 14], analysis of EMG burst characteristics (counts, magnitudes, durations and frequencies) [6, 9, 12, 13], analysis of EMG signal morphology [11], and methods of nonlinear EMG analysis [5]. It has been shown that there are differences in the burst characteristics between extension and flexion movements for patients with PD but not for neurologically healthy subjects [13]. In addition, it has been shown that the spiky morphology of PD-like EMG signals can be observed as a sharper EMG sample histogram and crossing rate (CR) expansion [11]. Furthermore, a nonlinear method of time-series analysis called the recurrence quantification analysis (RQA) has proven to be suitable in analyzing high degrees of MU synchronization in PD [5].

Previous studies where both EMG and acceleration signals have been analyzed have concentrated on analyzing the regularity of hand tremor and the low-frequency coherence in PD. It has been shown that the PD tremor is more regular (lower entropy) than the physiological tremor measured in neurologically healthy subjects [19]. In addition, it has been shown that the regularity of PD tremor reduces due to treatment of the disease by medication and deep brain stimulation [15]. The coherence studies on acceleration and EMG have shown that the low-frequency coherence between these signals is higher for patients with PD than for healthy persons [19]. Again, the medication and the deep brain stimulation reduce the pathological tremor-EMG coherence [15].

Measurements of EMG and limb movements can be used in the objective and quantitative assessment of the neuromuscular system function and motor disorders in PD. In the future, they may help in the diagnosis and follow-up of treatment in PD. In addition, neurophysiologic measurements may improve the understanding of the disease mechanisms [20]. However, these measurements are still rarely used in the assessment of PD. Because EMG is a spiky, impulse-like waveform, the nonlinear and morphological methods can be more effective in analyzing EMG than the methods that are traditionally used in the EMG analysis (amplitudes and mean/median frequencies).

In this paper, we present an advanced method for feature extraction and clustering of EMG and acceleration signals in PD. In the method, six different types of variables are calculated for both sides of the body and used to form high dimensional feature vectors. The dimensionality of these feature vectors is then reduced by using principal component approach. Finally, the cluster analysis of feature vectors is performed in a low-dimensional eigenspace. The variables used in this study were chosen for analysis based on their potential to characterize PD-like EMG signals. These variables are kurtosis, CR variable, correlation dimension and recurrence rate of EMG; sample entropy of acceleration; and coherence variable of EMG and acceleration. The method is tested with the data measured from 42 patients with PD (on-medication) and 59 healthy controls. The effect of PD medication is separately analyzed with a smaller number of patients ($N = 13$) that were measured both on- and off-medication.

2 Measurements

2.1 Subjects

Thirty-three healthy younger subjects (age 24 ± 8 years, height 174 ± 9 cm, weight 66 ± 12 kg), 26 healthy older subjects (age 61 ± 11 years, height 164 ± 8 cm, weight 72 ± 11 kg) and 42 patients with PD (age 65 ± 9 years, height 170 ± 9 cm, weight 75 ± 12 kg) participated in this study after giving their informed consent. The duration of PD was 8 ± 6 years (mean $\pm$ std) and UPDRS-motor score 27 ± 14 . The general inclusion criteria for patients were that they had an earlier diagnosis of idiopathic PD and they were taking medication for the condition. PD patients with other diseases that may interfere with motor function were excluded from the study. The control subjects were recruited from generally healthy Finnish and Russian citizens. This study was approved by the human ethics committees of Kuopio University Hospital and Petrozavodsk State University.

All 42 patients were measured on-medication because of ethical issues (inconvenient symptoms and tendency to fall). It has been shown previously, that the medication cannot restore the temporal structure of EMG signal into normal [12]. Thus, medication can reduce differences in the EMG signals between patients and healthy persons, but does not remove them completely. Although the cluster analysis of subjects was done with patients on-medication, we examined the effect of medication on the analysis results with a smaller number of patients (13 patients) who were also measured off-medication (no medication 24 h before the measurement). These patients were selected by a neurologist.

2.2 Measurement procedure

During the measurements, subjects were asked to hold their elbows at an angle of 90° with their palms up. The movement of the arm was not restricted. The isotonic contractions lasted for 10–30 s. No additional weights were used because the EMG bursts that are typical in PD are most visible in the signals of non-loaded condition [11]. In addition, the tremor is often reduced when the muscle becomes loaded. No loading was used to record EMG activities from control subjects either.

During the contraction, surface EMG was registered continuously from the biceps brachii (BB) muscles and the accelerations of forearms simultaneously from the wrists. Measurements were done by using ME6000-biosignal monitor (Mega Electronics Ltd., Kuopio, Finland). Both signals were sampled with a rate of 1,000 Hz.

Disposable Ag/AgCl electrodes (Medicotest, model M-00-S, Olstykke, Denmark) were used for EMG registration. They were attached bilaterally over the belly of the BB muscles with an inter-electrode spacing (center to center) of 3 cm. The inter-electrode spacing was chosen large enough to obtain information about the functioning of sufficient number of motor units. The reference electrodes were placed 6–7 cm laterally from the recording electrodes. Raw EMG signals were analogically band-pass filtered with an anti-aliasing filter (Butterworth, band-pass 1–500 Hz), amplified (differential amplifier, CMRR > 130 dB, total gain 1,000, noise < 1 µV) and analogue-to-digital converted (14-bit).

The accelerations of forearms were registered by using tri-axial accelerometers (Mega Electronics Ltd., range ±10 g, 14-bit AD converter), which were attached to palmar sides of subject's wrists with wrist supports. EMG and acceleration signals were stored in a PC for later analysis.

3 Data analysis

The following methods were used for feature extraction and clustering of EMG and acceleration signals. All data

were analyzed using Matlab™ (MathWorks Inc.) in five stages:

1. Pre-processing of the signals
2. Calculation of the EMG and acceleration variables
3. Formation of the feature vectors
4. Feature vector decomposition using the principal component approach
5. Cluster analysis of feature vectors in a low-dimensional eigenspace.

Bilateral EMG and acceleration signals of each subject were used for analysis and clustering.

3.1 Pre-processing of EMG and acceleration signals

The measured signals were first pre-processed to avoid the effects of noise and non-stationarities on EMG and acceleration variables and analysis results. The pre-processing included removal of low-frequency trends, exclusion of artifacts and division of signals into stationary overlapping epochs.

EMG signals were pre-processed as follows. First, 7-s long segments of EMG were chosen from the middle of the isotonic trial. It was visually verified that these EMG segments did not contain peaked artifacts caused by movement. That is, we did not use a low pass filter or other peak removal methods to remove peaked artifacts but we chose only artifact-free segments for analysis. Because the spikes and bursts that are typical for PD-like EMG signals occur regularly with a frequency between 3 and 12 Hz, the peaked artifacts were distinguished from them easily. The low-frequency trend (<10 Hz) was then removed from the signals by using a trend removal method called the smoothness priors method [16]. The trend removal could also be done by using regular high-pass filtering. Finally, the signal segments were divided into overlapping epochs of length 2,048 ms and 75% overlap.

Acceleration signals were pre-processed as follows. First, the simultaneous 7-s long segments of acceleration signals were chosen for analysis. The total acceleration was then calculated as a resultant of the three acceleration components. The trend was removed from the acceleration resultant by using the smoothness priors method (cut-off frequency ≈2 Hz). Finally, the acceleration signal was divided into overlapping epochs similarly with the corresponding EMG signal.

3.2 EMG and acceleration variables

Six different types of variables (i.e., features) were chosen for the analysis based on their potential to characterize PD-like EMG and acceleration signals. All variables were calculated for the EMG and acceleration signals from both

sides of the body. Therefore, the total number of variables was 12 (six right and six left side variables). Each variable was first calculated separately for the overlapping epochs and then averaged over the epochs. The variables used in this study were:

1. Kurtosis variable of EMG (k)
2. Crossing rate variable of EMG (CR)
3. Correlation dimension of EMG (D_2)
4. Recurrence rate of EMG (%REC)
5. Sample entropy of acceleration (SampEn)
6. Coherence variable of EMG and acceleration (Coh).

3.2.1 Kurtosis variable

The kurtosis variable is the fourth centered moment of a time series x

$$k = \frac{E\{(x - \mu)^4\}}{\sigma^4}, \quad (1)$$

where μ is the mean of the sample values and σ the standard deviation.

We calculated the kurtosis variable of EMG to assess the sharpness of the EMG sample distribution. Because the PD-like EMG signals typically are more spiky than the EMGs of healthy controls, the kurtosis variable was expected to be higher in PD.

3.2.2 Crossing rate variable

The CR variable was calculated for the EMG time series from the expansion of crossing rates. First, the CR expansion was formed by calculating the CRs at given threshold levels as described in [11]. The width of the expansion was then defined at the level of 50 crossings/s (level chosen in a pilot study) and the height as the maximum value of crossing rates at all threshold levels. Finally, the CR variable was calculated as the width/height of the expansion.

We calculated the CR variable of EMG to assess the spiky signal morphology in PD. Patients with PD were expected to have lower CR variables than the healthy persons because of the narrower CR expansions.

3.2.3 Correlation dimension

The correlation dimension assesses the complexity of biosignals. The calculation of it is based on reconstructing the biological system underlying the biosignal in a phase space. This can be done by applying Taken's embedding procedures as described for the neuromuscular system in [1]. If the system underlying the biosignal has a well-defined attractor (the asymptotic path of the

system) in the phase space, the dimension of this attractor is characterized by the correlation dimension [8]. We calculated the correlation dimension of EMG as described for R – R interval and systolic arterial pressure time series in [8]. However, the embedding parameters were chosen for EMG as delay = 5 and embedding dimension = 15.

3.2.4 Recurrence rate

The recurrence rate measures the percentage of recurring structures in a time series. The variable was introduced to analysis of physiological systems by Webber and Zbilut in 1994 [21]. As well as the calculation of the correlation dimension, the calculation of the recurrence rate is based on the system reconstruction in the phase space. We calculated the recurrence rate of EMG as described for EMG time series in [1]. Because of the increased level of MU synchronization, the PD-like EMG signals were expected to have higher recurrence rate than the EMGs of healthy persons.

3.2.5 Sample entropy

The sample entropy assesses the regularity of time series. The mathematical details about it are described in [10]. Briefly, the sample entropy measures the negative natural logarithm of the conditional probability that two sequences in a signal that are similar for m points are similar for $m + 1$ points. For totally random signal the sample entropy is near 2 and for pure sine wave near 0. We calculated the sample entropy of acceleration signals as described in [10]. Because the tremor is a typical symptom in PD, the regularity of acceleration signal was expected to be higher (lower sample entropy) for patients with PD than for healthy controls.

3.2.6 Coherence variable

Coherence describes the similarities in the power spectra of two time series. If the spectral estimates of the two time series are denoted with $P_x(f)$ and $P_y(f)$ and the cross-spectral estimate with $P_{xy}(f)$, the magnitude squared coherence estimate $C_{xy}(f)$ is

$$C_{xy}(f) = \frac{|P_{xy}(f)|^2}{P_x(f)P_y(f)}. \quad (2)$$

We estimated the coherence spectrum between EMG and acceleration signals by using the Welch's averaged periodogram method with a window length of 2,048 ms and 75% overlap. The magnitude squared coherence gives values between 0 and 1. The coherence variable was

calculated as the area of the coherence spectrum above a chosen threshold level (0.44) in the frequency band 0–50 Hz (band chosen in a pilot study).

3.3 Feature vectors

Feature vectors $\mathbf{z}$ were formed by using concatenated and normalized (to zero mean and unit standard deviation) EMG and acceleration variables

$$\mathbf{z} = [k_r, k_l, \%REC_r, \%REC_l, Coh_r, Coh_l, CR_r, CR_l, D_{2,r}, D_{2,l}, \text{SampEn}_r, \text{SampEn}_l]^T, \quad (3)$$

where the subscripts r and l denote the right or the left side of the body, respectively.

3.4 Principal component approach for feature vector decomposition

The dimensionality of the feature vectors was reduced by using the principal component approach. In this approach, the feature vector $\mathbf{z} \in \mathbb{R}^N$ for each subject is decomposed into orthogonal basis vectors $\phi_1, \dots, \phi_K$. That is, each feature vector is modeled as a weighted sum of these basis vectors. For the j 'th subject this sum is of the form

$$\mathbf{z}_j = \phi_1 \theta_j(1) + \phi_2 \theta_j(2) + \dots + \phi_K \theta_j(K) + \mathbf{v}_j, \quad (4)$$

where the scalar weights $\theta_j(1), \dots, \theta_j(K)$ are called the principal components and $\mathbf{v}_j$ is the model error.

The basis vectors and the principal components are solved for a group of feature vectors using the eigenvalue decomposition method. First, the feature vectors of M subjects are placed into the columns of a data matrix $\mathbf{Z}$

$$\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_M]. \quad (5)$$

After that, the data matrix is modeled with a linear model

$$\mathbf{Z} = \mathbf{H}\boldsymbol{\theta} + \mathbf{v}, \quad (6)$$

where $\mathbf{H} = [\phi_1, \dots, \phi_K] \in \mathbb{R}^{N \times K}$ is the model matrix, $\boldsymbol{\theta} = [\theta_1, \dots, \theta_M] \in \mathbb{R}^{K \times M}$ contains the weights and $\mathbf{v} = [\mathbf{v}_1, \dots, \mathbf{v}_M] \in \mathbb{R}^{N \times M}$ the model errors for all subjects. The basis vectors ϕ_k in $\mathbf{H}$ are selected to be the eigenvectors of the experimental correlation matrix

$$\mathbf{R} = \frac{1}{M} \sum_{j=1}^M \mathbf{z}_j \mathbf{z}_j^T = \frac{1}{M} \mathbf{Z} \mathbf{Z}^T. \quad (7)$$

It can be shown that by using the eigenvectors that correspond to K' ($K' < \min\{N, M\}$) largest eigenvalues the best K' -dimensional orthogonal approximation for the data set is obtained [7]. The principal components can be solved in the least squares sense from the linear model (Eq. 6) as

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{Z} = \mathbf{H}^T \mathbf{Z}, \quad (8)$$

where $\mathbf{H}^T \mathbf{H} = \mathbf{I}$ because the eigenvectors of $\mathbf{R}$ are orthonormal. In this study, the principal components are used for clustering of subjects.

3.5 Cluster analysis

The feature vectors of subjects were clustered into groups in a low-dimensional eigenspace by clustering the corresponding principal components. There are different methods available for cluster analysis. In this study, the main purpose of cluster analysis was not to introduce complex clustering tools but to show that there are differences in the EMG and acceleration signals between patients with PD and healthy persons. That is why we wanted to use a well-known and simple clustering algorithm.

In this study, we used the iterative k -means algorithm, which is one of the most popular clustering algorithms and it is conceptually and computationally simple [18]. The idea in k -means algorithm is as follows. The vectors to be clustered are denoted with p_j and the only parameter given to the algorithm is k , which is the number of clusters. The algorithm begins by choosing initial estimates for k cluster centroids. In each iteration step, it is determined to which cluster each vector p_j belongs. The vector p_j belongs to that cluster for which the squared Euclidean distance between the vector and the cluster centroid is minimized. In each iteration step, the cluster centroids are updated to be the mean of the vectors p_j in each cluster. The iteration continues until the sum of vector-to-centroid distances summed over all k clusters is minimized. In this study, we used a two-phase iterative algorithm. The first phase of it was explained above. In the second phase of the algorithm, vectors p_j are reassigned individually if doing so decreases the sum of distances. The cluster centroids are updated after each reassignment.

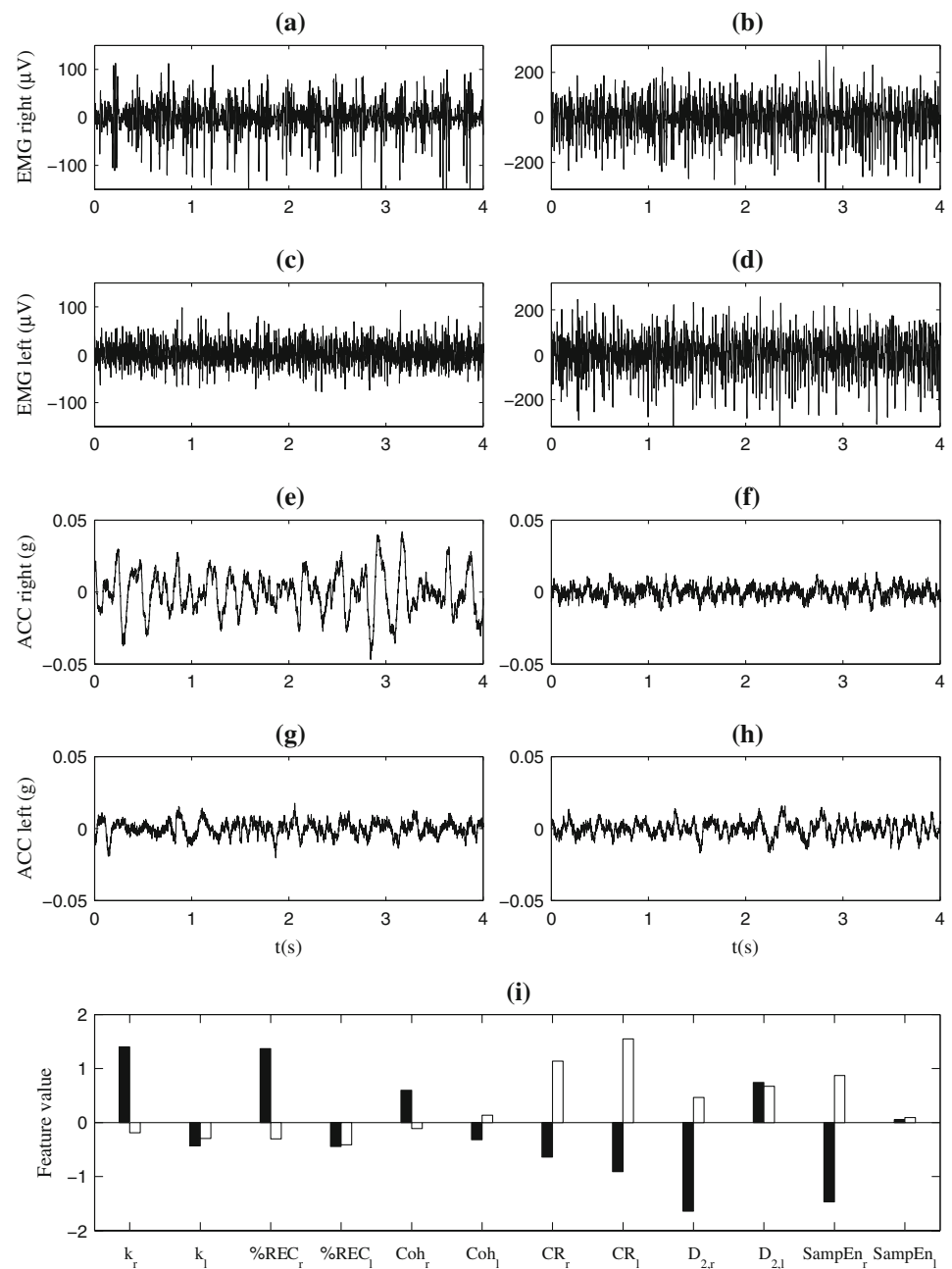
4 Results

4.1 Measured signals and feature vectors

Typical EMGs and acceleration signals measured from one patient with PD and one healthy person are presented in Fig. 1. The patient was more affected by the disease in the right side, which can be seen as a pattern of bursts in the right side EMG. In addition, the right side acceleration signal of the patient differs from the signals of the healthy person by having higher amplitude and being more regular.

The feature vector was formed for each subject by using normalized EMG and acceleration variables as described in Sect. 3.3. The normalized feature values for the selected

Fig. 1 **a** Right side EMG of one patient with PD. **b** Right side EMG of one healthy control. **c** Left side EMG of the patient. **d** Left side EMG of the healthy control. **e** Right side acceleration of the patient. **f** Right side acceleration of the healthy control. **g** Left side acceleration of the patient. **h** Left side acceleration of the healthy control. **i** Values of normalized features for the patient (*black*) and for the healthy control (*white*)



patient and healthy control are presented in Fig. 1. It is observed that the patient has higher values than the healthy control in the variables k_r , $\%REC_r$ and Coh_r . This means that the right side EMG of the patient is more spiky, more recurrent, and more coherent with the acceleration signal than the right side EMG of the healthy person. This was expected because the right side of the patient was more affected by the disease. In addition, the patient has lower values than the healthy control in the variables CR_r , CR_l , $D_{2,r}$ and $SampEn_r$. This means that the right side EMG and the acceleration signals of the patient are less complex than the right side signals of the healthy person. This is once

again explained by the side differences. In addition, CR variable is able to find differences between the patient and the control in the left side EMG.

4.2 Principal component approach

The feature vectors of all subjects (42 patients with PD on-medication, 33 healthy younger subjects, 26 healthy older subjects and 13 patients with PD off-medication) were set as column vectors to Eq. (5). The correlation matrix was then formed as defined in Eq. (7) and the eigenvalues and eigenvectors of it were solved. Seven eigenvectors

$(\phi_1, \dots, \phi_7)$ corresponding to the seven largest eigenvalues were chosen as the basis vectors. These seven eigenvectors contribute to 96% of the total variation in the feature vectors of all subjects. Thus, each feature vector can be quite accurately modeled as a weighted sum of these seven eigenvectors.

The principal components were then solved according to Eq. (8). The first, second and fifth principal component [$\theta_j(1)$, $\theta_j(2)$ and $\theta_j(5)$] were chosen for discrimination because they worked best in discriminating patients from healthy persons. The sum $\theta_j(2) + \theta_j(5)$ [later denoted with $\theta_j(y)$] worked even better for discrimination than the corresponding components alone. It should be noted that the order of the eigenvectors cannot be predicted before estimation. However, the eigenvectors that model relationships between feature vector components can be recognized by inspecting visually the morphology of the eigenvectors. In this study, we could recognize two things. First, the weights of the first eigenvector $\theta_j(1)$ describe the overall regularity of EMG and acceleration signals and the spiky morphology of EMG signals. Secondly, the weights of the second and fifth eigenvectors [$\theta_j(2)$ and $\theta_j(5)$] reflect differences between right and left side variables.

The feature vectors of subjects were clustered in a space spanned by $\theta_j(1)$ and $\theta_j(y)$. In that space (see Fig. 2), most of the healthy persons are located near the left border of the eigenspace presentation while the patients are more to the right. $\theta_j(1)$ is positive for most of the patients and negative for most of the healthy controls. The positive values of $\theta_j(1)$ indicate that the EMGs are more spiky, more recurrent and less complex (higher k and %REC and lower CR and D_2) than the EMGs with negative $\theta_j(1)$. In addition, it indicates that the acceleration signals are more regular (lower SampEn) and more coherent with the EMG signals.

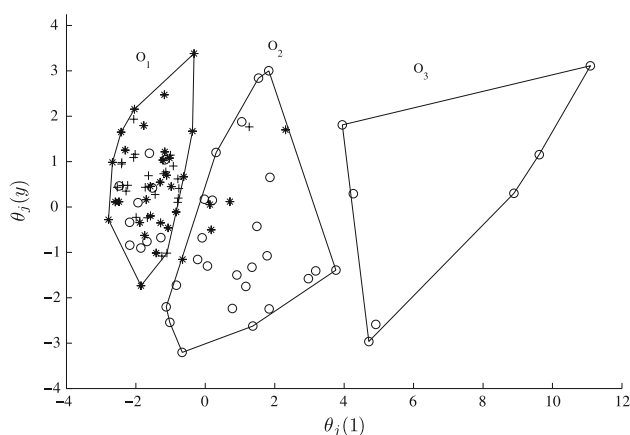


Fig. 2 Principal components $\theta_j(y)$ with respect to $\theta_j(1)$ of 42 patients with PD (circle), 33 healthy younger controls (asterisk) and 26 healthy older controls (plus). All the patients were measured on-medication. Subjects were clustered into three clusters O_1 , O_2 and O_3 by using k -means clustering algorithm

In the eigenspace presentation (Fig. 2), $\theta_j(y)$ is negative for most of the patients and positive for most of the controls. The negative values of $\theta_j(y)$ indicate that the right side EMG is more spiky, more regular and less complex than the left side EMG. In addition, it indicates that the right side acceleration is more regular and more coherent with the EMG signal than the left side acceleration. According to UPDRS-motor scores, most of the patients were more affected in the right side, which may explain the negative values of $\theta_j(y)$. In the eigenspace presentation, there is no clear distinction between healthy younger and older controls.

4.3 Cluster analysis

Cluster analysis of feature vectors was done in a low-dimensional eigenspace by using the k -means algorithm as described in Sect. 3.5. The number of patients, younger controls and older controls in each cluster (O_1 , O_2 and O_3) are detailed in Table 1. In addition, the total UPDRS-motor scores of patients (mean $\pm$ std) in each cluster are presented in Table 1.

According to clustering results, O_3 contains 17% of all patients and no healthy persons at all. The mean UPDRS-motor score of the patients in this cluster is relatively high. That is, the patients in this cluster have quite severe motor dysfunctions including tremor, rigidity and bradykinesia.

In O_2 , there are 60% of all patients. The mean UPDRS-motor score of patients in this cluster is lower than in O_3 . The patients in this cluster have similar symptoms than the patients in the third cluster, but the symptoms are milder. The second cluster contains six healthy controls of which five belong to the younger control group.

A total of 90% of all healthy persons in this study belong to O_1 . In addition, the first cluster contains 24% of all the patients. The mean UPDRS-motor score of patients in this cluster is relatively low. These patients have only little or no tremor at all in their hands. The most typical motor dysfunction of the patients in O_1 is the slowness of movements.

Table 1 Number of patients, younger controls and older controls in each cluster

Cluster	O_1	O_2	O_3
Number of patients ($N = 42$)	10	25	7
Number of younger controls ($N = 33$)	28	5	0
Number of older controls ($N = 26$)	25	1	0
Total number of subjects ($N = 101$)	63	31	7
Mean $\pm$ std UPDRS-motor score of patients	13 ± 9	26 ± 12	40 ± 11

Mean $\pm$ std values of UPDRS-motor scores in each cluster

The validation of the used discrimination method was performed as follows. The eigenvectors and principal components were computed for each combination of $M-1$ feature vectors, where M means the total number of subjects. That is, one subject was left out of the group each time the eigenvectors and principal components were computed. The clustering of feature vectors was then performed for each combination of $M-1$ subjects, and in each case, it was tested to which cluster the subject that was left out belongs. The correct ratings of clustering were defined as the percentage of controls that belonged to the control cluster and the percentage of patients that belonged to either of the two patient clusters. These ratings were $90 \pm 1\%$ and $74 \pm 6\%$ (mean $\pm$ std) for the controls and patients, respectively.

4.4 Effect of medication on feature vectors

The principal components were also calculated for 13 patients at off-medication state to demonstrate the effect of medication on feature vectors. In Fig. 3, the effect of medication is demonstrated in the subspace representation. One can see that most of the patients (10/13) are closer to the mean point of cluster O_1 when being on-medication than when being off-medication. That is, most patients are closer to healthy controls in this feature space when taking medication. Two of the other three patients are nearly in the same position when being on- and off-medication and one patient is clearly further from the healthy controls when taking medication.

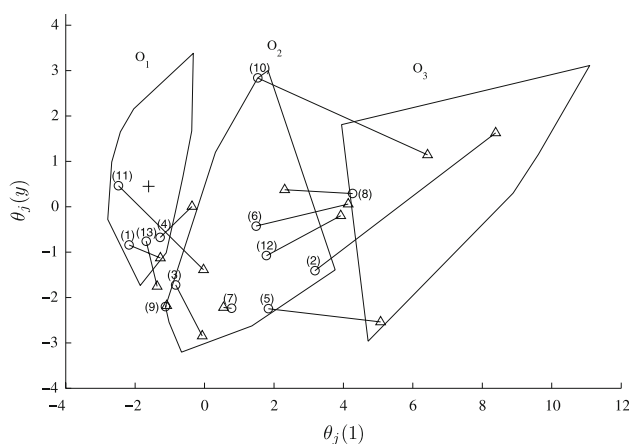


Fig. 3 Principal components $\theta_j(y)$ with respect to $\theta_j(1)$ of 13 patients with PD off-medication (triangle) and on-medication (circle). Patients are denoted with numbers (1)–(13) and the corresponding UPDRS-motor scores of them are in the off/on states: (1) 23/14, (2) 67/27, (3) 28/18, (4) 18/9, (5) 26/21, (6) 69/38, (7) 58/43, (8) 48/28, (9) 31/17, (10) 52/–, (11) 27/–, (12) 29/– and (13) 40/–. The off-medication and on-medication values are connected for each patient with a line. The previously defined three clusters are denoted with O_1 , O_2 and O_3 and the mean point of O_1 with plus symbol

5 Discussion

In this paper, an advanced method was presented for discrimination of surface EMG and acceleration signals between patients with PD and healthy persons. The method, which was tested with EMG and acceleration data measured from 42 patients and 59 healthy controls, resulted in rather good clustering of subjects. According to clustering results, one cluster contained 90% of the healthy controls and two other clusters 76% of the patients. Seven patients with severe motor dysfunctions were distinguished in one of the patient clusters.

The methods used previously in the research of EMG signals in PD can be divided into four categories: (1) spectral based methods [3, 5, 15, 19], (2) analysis of EMG burst characteristics during flexion and extension movements [6, 9, 12, 13], (3) morphology-based methods [11], and (4) nonlinear methods [5]. In this study, we did not examine flexion and extension movements but methods from the other three categories were applied for EMG feature extraction. In addition to EMG analysis, the regularity of acceleration signals was analyzed by calculating the sample entropy. It appeared that by forming high-dimensional feature vectors and by examining these feature vectors in a low-dimensional eigenspace, a good overall picture of the differences between PD-like and normal EMG and acceleration signals is obtained. The selected methods establish a promising approach for discrimination of EMG and acceleration signals between patients with PD and healthy persons.

The patients with PD in this study possessed various motor disorders (tremor, rigidity, bradykinesia and postural instability), not only tremor, and this was expected to affect the discrimination results. In fact, it appeared that the present approach is most sensitive to the tremor. In ideal situation, all the controls would have formed one cluster and all the patients other clusters. However, according to clustering results ten patients with only little or no tremor at all were clustered into the cluster of most healthy persons. The most typical symptom of these patients was slowness of movements. In the future, the discrimination results of these patients might be improved by analyzing EMG and acceleration signals during dynamic contractions as well. According to clustering results, there were patients with low UPDRS-motor scores that were discriminated from the control cluster by using the presented method. This indicates that the method could work in early stages of PD when the symptoms are mild. However, the sensitivity of this method in early diagnosis should be tested with a larger group of mildly impaired patients or with patients before actual diagnosis of the disease.

Five of the six incorrectly clustered healthy controls were younger (age < 45 years) persons and only one of

them was older (78 years). This indicates that ageing does not have similar effect on the EMG and acceleration features as PD has. The most typical feature for the incorrectly clustered healthy persons was spiky EMG signal morphology, which most likely resulted from an improper electrode placement, i.e., a large motor unit was firing constantly in the proximity of the recording electrode. If unchecked, also other non-Parkinsonian tremors (strong physiological tremor, essential tremor, metabolic tremor, etc.) may affect the EMG and acceleration signal features. Our control subjects did not, however, show tremor.

One might speculate that the clustering results could be improved by decreasing or increasing the number of variables in the feature vectors. The optimal set of variables was sorted out by gradually decreasing/increasing the number of variables in the approach. The 12 features used in this study were observed to describe essential EMG and acceleration characteristics during isotonic muscle contractions. Before this study, we performed the cluster analysis of feature vectors for each side of the body separately. In other words, we used six-element feature vectors containing only right or left side variables. However, the discrimination between patients and healthy persons was not so good in that case. This is probably due to the side differences that become visible in the second and fifth eigenvectors of the 12-element vector decomposition. Therefore, the information about side differences may enhance the discrimination power. If one would like to increase the discrimination power by using six-element feature vectors, that could be possible by selecting the more affected side of the patient for analysis. However, the more affected side was not known for all the patients in this study. After all, it must be noted that despite the side differences in patients with PD, the disease is not restricted to one side of the body only.

Comparison between the total UPDRS-motor scores and the principal components used in this study showed that there is a significant correlation (Spearman correlation, $p < 0.01$, $R = 0.53$) between the first principal component and the total UPDRS-motor score. However, this correlation was not considered in this study for two reasons. First, the total UPDRS-motor scores that are known about the patients do not directly measure the impairment in arm muscle function that was tested in this study. Secondly, the UPDRS examination was not performed at the same time with the measurement for all patients. In general, it is a well-known fact that the PD diagnosis can be difficult. Diagnostic accuracy by using subjective clinical evaluation is 70% in the early stages [17]. Imaging techniques can be of some help in the diagnosis. However, they are quite expensive and their differential specificity (68.5% for older patients) may not be very good in clinical practice [4]. We did not examine the sensitivity of our method in differential diagnosis. However, when thinking of these percentages,

EMG and acceleration measurements should be regarded as promising and cost-effective techniques for objective quantification of symptoms and signs in PD. Together with other already established techniques they may improve the diagnostic accuracy of PD.

Because of ethical reasons we could not examine all patients off-medication, since some of our patients may have an increased risk of falling. However, we examined the effect of medication on principal components with a smaller group of patients, which we considered “safe” as regards falling. It appeared that 10 of the 13 patients were closer to the cluster mean point of healthy controls when being on- than off-medication. This means that our present method for the discrimination of EMG and acceleration signals can detect medication-induced changes in the function of the neuromuscular system. The effect of medication on EMG and acceleration features was highly individual, which may describe the different states of the disease and different responses to medication (see Fig. 3). Large medication-induced changes in UPDRS-motor scores did not always result in large changes in EMG and acceleration features and vice versa. Two patients had nearly similar EMG and acceleration features when being on- and off-medication. One patient, that was clearly further from the healthy controls when being on- than off-medication, differed from other patients by taking anti-depressive medication. This may have altered the EMG and acceleration features analyzed in this study. Overall, the patient group that could be measured off-medication was not large enough to make significant conclusions. Further larger studies are suggested to evaluate the value of this presented method in the follow-up of PD medications and other treatments.

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